PROGRAMMING IN JAVA

Assignment 01

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which of the following is true regarding how Java handles code execution?

a. **Java uses only interpreter.**

b. **Java uses only compiler.**

c. **Java uses both interpreter and compiler.**

d. **None of the above.**

Correct Answer: c

c. **Java uses both interpreter and compiler.**

Detailed Solution:

Creating a .class file from .java using the javac command is a compilation task, whereas execution of a .class file using java is the process of interpretation.

QUESTION 2:

A Java file with extension '.class' contains:

a. **Java source code**

b. **HTML tags**

c. **Java Byte code**

d. **A program file written in Java programming language**

Correct Answer: c

c. **Java Byte code**

Detailed Solution:

A .class file is a compiled version of the .java file in byte code. It is a kind of object code with the JVM (Java Virtual Machine) as the target machine.

QUESTION 3:

Which of the following is NOT an object-oriented programming paradigm?

a. **Encapsulation**

b. **Inheritance**

c. **Polymorphism**

d. **Dynamic memory allocation**

Correct Answer: d

d. **Dynamic memory allocation**

Detailed Solution:

Encapsulation, Inheritance, and Polymorphism are core paradigms. Dynamic memory allocation is a memory allocation strategy and not a programming paradigm.

QUESTION 4:

What is the correct sequence of steps to execute a Java program?

a. **Run → Edit → Compile**

b. **Edit → Run → Compile**

c. **Compile → Edit → Run**

d. **Edit → Compile → Run**

Correct Answer: d

d. **Edit → Compile → Run**

Detailed Solution:

The Java development process involves writing code (Edit), converting it to bytecode (Compile), and then executing it on the JVM (Run).

QUESTION 5:

Which program is used to compile Java source code into bytecode?

a. **javap**

b. **javac**

c. **java**

d. **javad**

Correct Answer: b

b. **javac**

Detailed Solution:

The JDK includes a tool called 'javac' that compiles from Java source code to a target of Java bytecodes. The program 'java' is used to execute Java bytecode.

QUESTION 6:

In Java, what is the role of the public static void main(String[] args) method?

a. **Initialization method**

b. **Execution entry point**

c. **Constructor**

d. **Destructor**

Correct Answer: b

b. **Execution entry point**

Detailed Solution:

The public static void main(String[] args) method is the entry point for the execution of a Java program.

QUESTION 7:

Which of the following is not a feature of Java?

a. **Platform Independence**

b. **Object-Oriented Programming**

c. **Supports Explicit Pointers**

d. **Supports Polymorphism**

Correct Answer: c

c. **Supports Explicit Pointers**

Detailed Solution:

Java is platform-independent, object-oriented, and secure. It does not support explicit pointers, as they can lead to memory management issues and compromise security.

QUESTION 8:

What is the primary purpose of the Java Virtual Machine (JVM) in the Java programming language?

a. **Graphical User Interface**

b. **Platform independence**

c. **Code Debugging**

d. **Game Engine**

Correct Answer: b

b. **Platform independence**

Detailed Solution:

The Java Virtual Machine (JVM) enables platform independence by interpreting Java bytecode, allowing Java programs to run on any device with a compatible JVM.

QUESTION 9:

Consider the following program.

```
public class Question {
    public static void main(String[] args) {
        int x = 5;
        x *= (2 + 8);
        System.out.println(x);
    }
}
```

What is the output of the above code?

a. **50**

b. **10**

c. **Compiler error**

d. **5**

Correct Answer: a

a. **50**

Detailed Solution:

Here, x *= (2 + 8) is equivalent to x = x * (2 + 8), which results in x = x * 10. Therefore, x = 5 * 10 => x = 50.

QUESTION 10:

Consider the following program.

```
class NPTEL {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

What is the output of the above code?

a. **Hello, World!**

b. **HelloWorld!**

c. **Compilation Error**

d. **Runtime Error**

Correct Answer: a

a. **Hello, World!**

Detailed Solution:

This is the standard syntax for a Java program to print the string "Hello, World!" to the console.

PROGRAMMING IN JAVA

Assignment 02

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

How do you read a line of text from the console using the Scanner class in Java?

a. **scanner.readLine()**

b. **scanner.nextLine()**

c. **scanner.getLine()**

d. **scanner.fetchLine()**

Correct Answer: b

b. **scanner.nextLine()**

Detailed Solution:

The nextLine() method of the Scanner class reads a line of text from the console. Refer to Lecture 10 for more details.

QUESTION 2:

What will be the output of the following Java program?

```
public class VarPrint {
    int x = 30;
    static int y = 20;

    public static void main(String[] args) {
        VarPrint t1 = new VarPrint();
        t1.x = 99;
        t1.y = 88;
        int z1 = t1.x + t1.y;
        VarPrint t2 = new VarPrint();
        System.out.println(t2.x + " " + t2.y + " " + z1);
    }
}
```

a. **30 99 178**

b. **30 88 129**

c. **30 99 187**

d. **30 88 187**

Correct Answer: d

d. **30 88 187**

Detailed Solution:

If you perform any change for instance variable these changes won't be reflected for the remaining objects. Because for every object a separate copy of instance variable will be there. But if you do any change to the static variable, that change will be reflected for all objects because a static instance maintains a single copy in memory.

Please refer to chapter 3 of book Joy With Java for a more detailed explaination.

QUESTION 3:

What will be the output of the following Java program?

```
public class ArgumenTest {
    public static void main(String[] args) {
        Test t = new Test();
        t.start();
    }

    static class Test {
        void start() {
            int a = 4;
            int b = 5;
            System.out.print("" + 8 + 3 + "");
            System.out.print(a + b);
            System.out.print(" " + a + b + "");
            System.out.print( a + b + "");
            System.out.print(foo() + a + b + " ");
            System.out.println(a + b + foo());
        }

        String foo() {
            return "foo";
        }
    }
}
```

a. **9 7 7 foo34 34foo**

b. **839 45foo45 9foo**

c. **839 459foo45 9foo**

d. **9 7 7 foo 7 7foo**

Correct Answer: c

c. **839 459foo45 9foo**

Detailed Solution:

Here, print() methods internally converts the data in its argument into a String object and then print the composition. Here, + is the concatenation of different String representation.

Please refer to chapter 3 of book Joy With Java for a more detailed explaination.

QUESTION 4:

What is encapsulation in object-oriented programming?

a. **Hiding implementation details and exposing only functionality**

b. **The process of creating multiple objects in a program**

c. **Writing multiple methods in a single class**

d. **Using the this keyword to reference an object**

Correct Answer: a

a. **Hiding implementation details and exposing only functionality**

Detailed Solution:

Please refer to chapter 3 of book Joy With Java for a more detailed explaination.
Encapsulation is one of the fundamental principles of object-oriented programming. It involves bundling the data (variables) and the methods (functions) that operate on the data into a single unit, typically a class. By making certain data private and providing public methods to access or modify it, encapsulation helps hide implementation details from the outside world while exposing only the required functionality. This improves code modularity, security, and maintainability.

QUESTION 5:

Consider the following code snippet

```
class NPTEL_W2 {
  int x;

  NPTEL_W2(int x) {
      this.x = x+1;
  }

  void printX() {
      System.out.println(this.x);
  }

  public static void main(String[] args) {
      NPTEL_W2 obj = new NPTEL_W2(100);
      obj.printX();
  }
}
```

What will be the output of the code given above?

a. **0**

b. **101**

c. **100**

d. **Runtime error**

Correct Answer: b

b. **101**

Detailed Solution:

The constructor `NPTEL_W2 (int x)` initializes the instance variable `x` with the value passed as an argument. The method `printX()` prints the value of `x`, which is `101`. Refer to Lecture 7 for more details.

QUESTION 6:

What does the `this` keyword in Java help to achieve?

a. **Avoiding name space collision between instance variables and method parameters**

b. **Overloading methods in a class**

c. **Accessing private methods in another class**

d. **Creating multiple objects in a program**

Correct Answer: a

a. **Avoiding name space collision between instance variables and method parameters**

Detailed Solution:

The *this* keyword is used to refer to the current instance of a class. It is commonly used to resolve ambiguity when instance variables and method parameters have the same name. For example:

```
class Example {
    int value;

    Example(int value) {
        this.value = value; // Resolves name collision by referring to
the instance variable
    }
}
```

Here, this.value refers to the instance variable, while value refers to the parameter of the constructor. This avoids confusion and ensures the proper assignment of values.
Please refer book Joy with Java Chapter 3 for mored etailed explaination.

QUESTION 7:

Which method is used as the starting point of execution in a Java program?

a. **init method**

b. **start method**

c. **main method**

d. **run method**

Correct Answer: c

c. **main method**

Detailed Solution:

You are encouraged to review the role of program execution flow in Java on your own.
Note: Please refer to the book Joy with Java Chapter 3 for more detailed explanation.

QUESTION 8:

Which of the following is used to display output in Java?

a. **input**

b. **display**

c. **print**

d. **System**

Correct Answer: c

c. **print**

Detailed Solution:

The print method is used to display output on the screen in Java.
It is available through the System.out object.
You should practice identifying commonly used output statements.
Note: Please refer to the book Joy with Java Chapter 3 for more detailed explanation.

QUESTION 9:

Which symbol is used to end a statement in Java?

a. :

b. .

c. ;

d. ,

Correct Answer: c

c. ;

Detailed Solution:

In Java, every statement must end with a semicolon.
It tells the compiler that the statement is complete.
Revise Java syntax rules carefully.
Refer to Joy with Java Chapter 2 for clarification.

QUESTION 10:

Which of the following is a valid Java keyword?

a. **number**

b. **class**

c. **include**

d. **define**

Correct Answer: b

b. **class**

Detailed Solution:

The class keyword is used to define a class in Java.
Java keywords have predefined meanings.
Students should memorize basic Java keywords.

Note: Please refer to chapter 3 of book Joy With Java for a more detailed explaination.

PROGRAMMING IN JAVA

Assignment 03

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which of the following statements is true regarding the order of execution of constructors in an inheritance hierarchy?

a. **Derived class constructor will be called followed by the base class constructor.**

b. **Only Base class constructor will be called.**

c. **Only derived class constructor will be called.**

d. **Base class constructor will be called followed by the derived class constructor.**

Correct Answer: d

d. **Base class constructor will be called followed by the derived class constructor.**

Detailed Solution:

On object creation of a derived class, Java ensures that the base (parent) class constructor is executed first to initialize the parent's state before executing the derived class constructor.

QUESTION 2:

The super() method is used to:

a. **Call constructor of friend class**

b. **Call constructor of the parent class**

c. **Call constructor**

d. **Is a declared method**

Correct Answer: b

b. **Call constructor of the parent class**

Detailed Solution:

In Java, `super()` is used to invoke the constructor of the immediate parent class. It allows the subclass to initialize the parent class's state.

QUESTION 3:

What is the output of the following Java program?

```
class Vehicle {
    public void move() {
        System.out.println("The vehicle moves");
    }
}

class Car extends Vehicle {
    public void move() {
        System.out.println("The car moves");
    }
}

public class Main {
    public static void main(String[] args) {
        Vehicle vehicle = new Car();
        vehicle.move();
    }
}
```

a. **"The vehicle moves"**

b. **"The car moves"**

c. **The code does not compile**

d. **None of the above**

Correct Answer: b

b. **"The car moves"**

Detailed Solution:

This is an example of method overriding. Since the object is instantiated as a `Car`, the overridden `move` method in the `Car` class is executed, even though the reference type is `Vehicle`.

QUESTION 4:

Which of the following is NOT true regarding the final keyword in Java?

a. **A final method cannot be overridden in a subclass.**

b. **A final variable can only be assigned once.**

c. **A final class can have subclasses.**

d. **A final variable can be assigned during declaration or in the constructor.**

Correct Answer: c

c. **A final class can have subclasses.**

Detailed Solution:

A class marked as `final` cannot be extended, meaning it cannot be subclassed (it cannot have any child classes).

QUESTION 5:

Which type of inheritance leads to the "Diamond Problem"?

a. **Single level**

b. **Multi-level**

c. **Multiple**

d. **Hierarchical**

Correct Answer: c

c. **Multiple**

Detailed Solution:

The diamond problem arises in multiple inheritance when a class inherits from two classes that have a common ancestor, leading to ambiguity regarding which path to follow for inherited members. Java avoids this by not supporting multiple inheritance with classes.

QUESTION 6:

Members of a class which are not intended to be inherited by subclasses should be declared as:

a. **Public members**

b. **Protected members**

c. **Private members**

d. **Private or Protected members**

Correct Answer: c

c. **Private members**

Detailed Solution:

The `private` access specifier is the most restrictive. Private members are not visible to subclasses and therefore are not inherited.

QUESTION 7:

In Java, is it possible to override a static method?

a. **Yes, we can override a static method just like we do with instance methods.**

b. **No, static methods cannot be overridden because they belong to the class, not the object.**

c. **It depends on whether the static method is declared as final or not.**

d. **It depends on the access modifier of the static method.**

Correct Answer: b

b. **No, static methods cannot be overridden because they belong to the class, not the object.**

Detailed Solution:

Static methods are bound to the class rather than the instance (static binding). If a subclass defines a static method with the same signature, it "hides" the parent method rather than overriding it.

QUESTION 8:

What is the output of the following Java program?

```
public class NptelExample {
    public static int fun(int n) {
        if (n == 0) {
            return 1;
        }
        return n * fun(n - 1);
    }

    public static void main(String[] args) {
        System.out.println(fun(5));
    }
}
```

a. 5

b. 24

c. 120

d. Runtime Error

Correct Answer: c

c. 120

Detailed Solution:

The `fun` method is a recursive function calculating the factorial of `n`. For input 5, it calculates 5 * 4 * 3 * 2 * 1 = 120.

QUESTION 9:

Which of the following statements about abstract classes in Java is correct?

a. **Abstract classes can be instantiated directly.**

b. **An abstract class must contain at least one abstract method.**

c. **A class inheriting from an abstract class must implement all its abstract methods unless it is itself abstract.**

d. **Abstract classes can be marked as final.**

Correct Answer: c

c. **A class inheriting from an abstract class must implement all its abstract methods unless it is itself abstract.**

Detailed Solution:

Abstract classes cannot be instantiated. Any concrete subclass (non-abstract) must provide implementations for all abstract methods defined in the parent abstract class.

QUESTION 10:

What will be the output of the below program?

```
class superDemoClass {
    final int a = 20;
}

class subDemoClass extends superDemoClass {
    void subFunc() {
        a = 40;
        System.out.println("value of a = " + a);
    }
}

class demo {
    public static void main(String[] args) {
        subDemoClass subc = new subDemoClass();
        subc.subFunc();
    }
}
```

a. **value of a = 20**

b. **error: cannot assign a value to final variable a**

c. **error: unknown variable 'a' in class subDemoClass**

d. **value of a = 40**

Correct Answer: b

b. **error: cannot assign a value to final variable a**

Detailed Solution:

The variable `a` is declared as `final` in the superclass. While it is inherited, its value cannot be modified (re-assigned) by the subclass, causing a compilation error.

PROGRAMMING IN JAVA

Assignment 4

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which access modifier allows access only inside the same package?

a. **default**

b. **private**

c. **public**

d. **protected**

Correct Answer: a

a. **default**

Detailed Solution:

The default access modifier allows access within the same package only.
It is applied when no access modifier is written.
Revise package-level access rules carefully.

QUESTION 2:

Which constructor is called first when an object of a child class is created? The method will be accessible.

a. **Child class constructor**

b. **Parent class constructor**

c. **Interface constructor**

d. **Package constructor**

Correct Answer: b

b. **Parent class constructor**

Detailed Solution:

The parent class constructor is invoked first automatically.
This happens through an implicit call to super().
Revisit inheritance basics and constructor order.

QUESTION 3:

What will be the output of the following code?

```
class Person {
    int a = 1;
    int b = 0;
    public Person() {
        System.out.println(a + b + " Java " );
    }
}

class Employee extends Person {
    int a = 0;
    int b = 1;
    public Employee() {
        System.out.println(""+a + b + " Java " );
    }
}

public class Question {
    public static void main(String args[]) {
        Person p = new Employee();
    }
}
```

a. **1 Java 10**
   0 Java 0

b. **1 Java**
   0 Java

c. **1 Java**
   01 Java

d. **1 java 1**
   0 Java 0

Correct Answer: c

c. **1 Java**
   01 Java

Detailed Solution:

If no super() or this() is included explicitly within the derived class constructor, the super() is implicitly invoked by the compiler. Therefore, in this case, the Person class constructor is called first and then the Employee class constructor is called. Up casting is allowed.

QUESTION 4:

Which statement is used to use a class from another package?

a. **package**

b. **include**

c. **import**

d. **extends**

Correct Answer: c

c. **import**

Detailed Solution:

The import statement allows access to classes from other packages.
It improves code readability and reuse.
Practice basic import statements.

QUESTION 5:

What keyword is used to define a package in Java?

a. **define**

b. **package**

c. **import**

d. **namespace**

Correct Answer: b

b. **package**

Detailed Solution:

The `package` keyword is used in Java to define a package. For example, `package mypackage;` defines a package named `mypackage`.

QUESTION 6:

How do you import a specific class from a package in Java?

a. **import package.*;**

b. **import package.ClassName;**

c. **include package.ClassName;**

d. **use package.ClassName;**

Correct Answer: b

b. **import package.ClassName;**

Detailed Solution:

The syntax `import package.ClassName;` imports a specific class from a package. For example, `import java.util.Scanner;` imports the `Scanner` class from the `java.util` package.

QUESTION 7:

Consider the following 2 programs:

```java
// Main1.java ------------------
public class Main1{
    public static void main(String args[]){
        int number = 10;
        System.out.println(number++ + ++number);
    }
}
```

```java
// Main2.java ------------------
public class Main2{
    public static void main(String args[]){
        int number = 10;
        System.out.println(++number + number++);
    }
}
```

Choose the best option among the following for the code snippet given above

a. **Both pre-increment and post-increment operators becomes pre-increment during print.**

b. **Both pre-increment and post-increment operators becomes post-increment during print.**

c. **Both Main1 and Main2 classes give the same output.**

d. **Pre-increment and post-increment operators don't work during print.**

Correct Answer: c

c. **Both Main1 and Main2 classes give the same output.**

Detailed Solution:

The output of both the program are 22. Therefore, option **c** is correct and we can eliminate option **d** that the operators don't work. Further, the operators are doing exactly what they are supposed to do i.e. pre-increment first increases the values and post-increment increases the value during the next operation. The print statement is the next operation; hence it received the post incremented value as well making option **a** and **b** invalid.

QUESTION 8:

Consider the code given below.

```
// Teacher.java ------------------
package nptel1;
public class Teacher {
  protected void showMarks() {
    System.out.println("100 Marks");
  }
}
```

```
// Student.java ------------------
package nptel2;
import nptel1.*;
public class Student extends Teacher {
  void show() {
    showMarks();
  }
  public static void main(String[] args) {
    Student st1 = new Student();
    st1.show();
  }
}
```

What is the output of the above Java Code Snippet with `protected` access modifier?

a. **100 Marks**

b. **No output**

c. **Compiler error**

d. **None of the above**

Correct Answer: a

a. **100 Marks**

Detailed Solution:

Through inheritance, one can access a `protected` variable or method of a class even from outside the package. Here, we accessed `Teacher` class of `nptel1` from `Student` class of `nptel2`.

QUESTION 9:

What will happen if you do not specify an access modifier for a class in a package?

a. **The class will be `private` by default.**

b. **The class will be `protected` by default.**

c. **The class will be accessible only within the same package.**

d. **The class will be `public` by default.**

Correct Answer: c

c. The class will be accessible only within the same package.

Detailed Solution:

If no access modifier is specified for a class, it gets the default access level, making it accessible only within the same package.

QUESTION 10:

Which access modifier provides the most restrictive access in Java?

a. **default**

b. **protected**

c. **private**

d. **public**

Correct Answer: c

c. **private**

Detailed Solution:

The `private` modifier provides the most restrictive access, allowing access only within the class where it is defined.

PROGRAMMING IN JAVA

Assignment 5

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which keyword is NOT used by Java during exception handling?

a. **try**

b. **catch**

c. **final**

d. **finally**

Correct Answer: c

c. **final**

Detailed Solution:

In Java, exceptions are handled using the `try`, `catch`, and `finally` blocks. The `try` block contains code that might throw an exception, the `catch` block handles specific exceptions, and the `finally` block executes regardless of whether an exception occurs.

QUESTION 2:

What is the purpose of the `throws` keyword in Java?

a. **To declare exceptions that a method can throw**

b. **To throw an exception immediately**

c. **To catch an exception**

d. **It is not a keyword in Java**

Correct Answer: a

a. **To declare exceptions that a method can throw**

Detailed Solution:

The `throws` keyword is used in a method signature to declare the exceptions that the method might throw, alerting callers to handle or propagate these exceptions.

QUESTION 3:

Which of the following is TRUE about interfaces in Java?

a. **Interfaces should always be defined as final**

b. **Interfaces can be instantiated directly.**

c. **Interfaces can extend multiple interfaces.**

d. **Interfaces cannot have any method signatures**

Correct Answer: c

c. **Interfaces can extend multiple interfaces.**

Detailed Solution:

In Java, an interface can extend multiple interfaces. Interfaces can also contain `public`, `static`, and `final` variables, but they cannot be instantiated directly.

QUESTION 4:

What will be the output of the following code?

```
interface Demo {
    void display();
}

class Test implements Demo {
    public void display() {
        System.out.println("Hello, NPTEL!");
    }
}

public class Main {
    public static void main(String[] args) {
        Test obj = new Test();
        obj.display();
    }
}
```

a. **Hello, NPTEL!**

b. **Compilation Error**

c. **Runtime Error**

d. **No Output**

Correct Answer: a

a. **Hello, NPTEL!**

Detailed Solution:

The `Test` class implements the `Demo` interface and provides a definition for the `display` method. When `display()` is called, it prints “Hello, NPTEL!”.

QUESTION 5:

What will be the output of the following Java program?

```
interface Calculator {
    void calculate(int value);
}

class Square implements Calculator {
    int result;
    public void calculate(int value) {
        result = value * value;
        System.out.print("Square: " + result + " ");
    }
}

class Cube extends Square {
    public void calculate(int value) {
        result = value * value * value;
        super.calculate(value);
        System.out.print("Cube: " + result + " ");
    }
}

public class Main {
    public static void main(String[] args) {
        Calculator obj = new Cube();
        obj.calculate(3);
    }
}
```

a. **Square: 9 Cube: 9**

b. **Cube: 27 Square: 9**

c. **Square: 9 Square: 27 Cube: 27**

d. **Square: 9 Cube: 27 Square: 27**

Correct Answer: a

a. **Square: 9 Cube: 9**

Detailed Solution:

The `Cube` class overrides the `calculate` method of the `Square` class. In the `Cube` class's `calculate` method, `super.calculate(value)` is called, which executes the `calculate` method of the `Square`

class. First, "Square: 9" is printed by the superclass method. The call to super.calculate(value) overwrites the previously computed cube value. Then, the overridden method in `Cube` prints "Cube: 9".

QUESTION 6:

What will be the output of the following program?

```
class NPTEL {
    public static void main(String[] args) {
        try {
            int a = 5;
            int b = 0;
            System.out.println(a / b);
        } catch (ArithmeticException e) {
            System.out.print("Error ");
        } finally {
            System.out.print("Complete");
        }
    }
}
```

a. **5 Complete**

b. **Error Complete**

c. **Runtime Error**

d. **Compilation Error**

Correct Answer: b

b. **Error Complete**

Detailed Solution:

An `ArithmeticException` is caught in the `catch` block, which prints “Error”. The `finally` block executes afterward, printing “Complete”.

QUESTION 7:

What will be the output of the following Java program?

```
interface A {
    int x = 10;
    void display();
}
class B implements A {
    public void display() {
        System.out.println("Value of x: " + x);
    }
}
public class Main {
    public static void main(String[] args) {
        B obj = new B();
        obj.display();
    }
}
```

a. **Value of x: 10**

b. **Value of x: 0**

c. **Compilation Error**

d. **Runtime Error**

Correct Answer: a

a. **Value of x: 10**

Detailed Solution:

Variables in interfaces are `public`, `static`, and `final` by default. Hence, the value of `x` is accessible in the `display` method.

QUESTION 8:

Which keyword is used by Java during exception handling?

a. **final**

b. **finally**

c. **caught**

d. **finished**

Correct Answer: b

b. **finally**

Detailed Solution:

In Java, exceptions are handled using the try, catch, and finally blocks. The try block contains code that might throw an exception, the catch block handles specific exceptions, and the finally block executes regardless of whether an exception occurs.

QUESTION 9:

A method that potentially generates a checked exception must include this keyword in its method signature:

a. **throw**

b. **extend**

c. **throws**

d. **extends**

Correct Answer: c

c. **throws**

Detailed Solution:

Any Java class that generates a checked exception and does not handle it internally must use the throws keyword to alert other methods of its instability.

QUESTION 10:

Consider the following code.

```
class Test extends Exception {}

class Main {

  public static void main(String args[]) {
    try {
      throw new Test();
    } catch (Test t) {
      System.out.println("Got the Test Exception");
    } finally {
      System.out.println("Inside finally block ");
    }
  }
}
```

What will be the output of the above code?

a. **Got the Test Exception**
   Inside finally block

b. **Got the Test Exception**

c. **Inside finally block**

d. **Compiler Error**

Correct Answer: a

a. **Got the Test Exception**
   Inside finally block

Detailed Solution:

In Java, the finally is always executed after the try-catch block. This block can be used to do the common cleanup work.

PROGRAMMING IN JAVA

Assignment 06

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which method is used to start a new thread in Java?

a. `run()`

b. `start()`

c. `execute()`

d. `begin()`

Correct Answer: b

b. start()

Detailed Solution:

Calling `start()` creates a new thread and internally invokes the `run()` method. Calling `run()` directly does not create a new thread.

QUESTION 2:

Which interface must be implemented to create a thread using the Runnable approach?

a. `Threadable`

b. `Runnable`

c. `Callable`

d. `Executor`

Correct Answer: b

b. Runnable

Detailed Solution:

The `Runnable` interface contains a single method `run()` and is used when a class needs to be executed by a thread.

QUESTION 3:

What happens if a thread's `run()` method completes execution?

a. **The thread enters the waiting state**

b. **The thread can be restarted**

c. **The thread enters the terminated state**

d. **The thread returns to the runnable state**

Correct Answer: c

c. **The thread enters the terminated state**

Detailed Solution:

Once the `run()` method finishes, the thread completes its lifecycle and enters the terminated state. A terminated thread cannot be restarted.

QUESTION 4:

Which keyword is used to synchronize a method in Java?

a. `Lock`

b. `Threadsafe`

c. `Synchronized`

d. `Volatile`

Correct Answer: c

c. Synchronized

Detailed Solution:

The `synchronized` keyword ensures that only one thread can access a method or block at a time, preventing race conditions.

QUESTION 5:

Which of the following is the default priority of a Java thread?

a. 1

b. 3

c. 5

d. 2

Correct Answer: c

c. 5

Detailed Solution:

Java thread priorities range from 1 (`MIN_PRIORITY`) to 10 (`MAX_PRIORITY`). The default priority is 5 (`NORM_PRIORITY`).

QUESTION 6:

Which method causes the currently executing thread to pause for a specified time?

a. `wait()`

b. `sleep()`

c. `pause()`

d. `delay()`

Correct Answer: b

b. sleep()

Detailed Solution:

The `sleep()` method pauses the currently executing thread for a specified number of milliseconds without releasing any locks.

QUESTION 7:

Which exception is thrown when `sleep()` is interrupted?

a. **RuntimeException**

b. **IOException**

c. **InterruptedException**

d. **ThreadException**

Correct Answer: c

c. **InterruptedException**

Detailed Solution:

`sleep()` throws `InterruptedException` if another thread interrupts the sleeping thread.

QUESTION 8:

Which class is used to read text data efficiently from a file in Java?

a. `FileInputStream`

b. `BufferedReader`

c. `DataInputStream`

d. `ObjectInputStream`

Correct Answer: b

b. BufferedReader

Detailed Solution:

`BufferedReader` reads text efficiently using buffering and provides the `readLine()` method for line-by-line input.

QUESTION 9:

Which method is used to read a line of text using `BufferedReader`?

a. `read()`

b. `readText()`

c. `readLine()`

d. `getLine()`

Correct Answer:

c. readLine()

Detailed Solution:

The `readLine()` method reads an entire line of text and returns it as a `String`. It returns `null` at end of file.

QUESTION 10:

Which class is the superclass of all input byte streams in Java?

a. **Reader**

b. **InputStream**

c. **FileInputStream**

d. **BufferedInputStream**

Correct Answer:

b. **InputStream**

Detailed Solution:

`InputStream` is the abstract superclass for all byte-based input streams in Java, such as `FileInputStream`.

PROGRAMMING IN JAVA

Assignment VII

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

What will be the output of the following Java program?

```
import java.io.*;
class ReadFile {
    public static void main(String[] args) throws IOException {
        FileReader fr = new FileReader("NPTEL.txt");
        BufferedReader br = new BufferedReader(fr);
        System.out.println(br.readLine());
        br.close();
    }
}
```

Assume `NPTEL.txt` contains:

```
This is Programming in Java online course.
```

a. **Hello, World!**

b. **This is Programming in Java online course.**

c. **IOException**

d. **null**

Correct Answer: b

b. **This is Programming in Java online course.**

Detailed Solution:

`BufferedReader` to read the first line from the file `NPTEL.txt`. Since the file contains “This is Programming in Java online course.”, it is printed.

QUESTION 2:

Complete the following snippet with the required code.

```
File file = new File("file.txt");
if (____________________) // Fill in the blanks
{
    System.out.println("File exists.");
} else
{
    System.out.println("File does not exist.");
}
```

a. **file.exists()**

b. **file.isFile()**

c. **file.fileExists()**

d. **file.isAvailable()**

Correct Answer: a

a. **file.exists()**

Detailed Solution:

The `exists()` method in the `File` class checks if a file or directory exists in the specified path.

QUESTION 3:

What is the output of the following Java program?

```
class Main {

  public static void main(String args[]) {
    final int i;
    i = 20;
    i = 30;
    System.out.println(i);
  }
}
```

a. 30

b. Compiler Error

c. Garbage value

d. 0

Correct Answer: b

b. Compiler Error

Detailed Solution:

i is assigned a value twice. Final variables can be assigned values only one. Following is the compiler error “Main.java:5: error: variable i might already have been assigned”.

QUESTION 4:

Which of these values is returned by read() method when the end of file (EOF) is encountered?

a. 0

b. 1

c. -1

d. Null

Correct Answer: c

c. -1

Detailed Solution:

Each time read() is called, it reads a single byte from the file and returns the byte as an integer value. read() returns -1 when the end of the file is encountered.

QUESTION 5:

What will be the output of the following Java program?

```
public class Calculator {
    int num = 100;

    public void calc(int num) {
        this.num = num * 10;
    }

    public void printNum() {
        System.out.println(num);
    }

    public static void main(String[] args) {
        Calculator obj = new Calculator();
        obj.calc(2);
        obj.printNum();
    }
}
```

a. 20

b. 100

c. 1000

d. 2

Correct Answer: a

a. 20

Detailed Solution:

Here the class instance variable name(num) is same as calc() method local variable name(num). So for referencing class instance variable from calc() method, this keyword is used. So in statement this.num = num * 10, num represents local variable of the method whose value is 2 and this.num represents class instance variable whose initial value is 100. Now in printNum() method, as it has no local variable whose name is same as class instance variable, so we can directly use num to reference instance variable, although this.num can be used.

QUESTION 6:

What is the primary difference between Byte Streams and Character Streams in Java?

a. **Byte Streams handle characters, while Character Streams handle bytes.**

b. **Byte Streams are used for binary data, while Character Streams are used for image data.**

c. **Character Streams are faster than Byte Streams in all cases.**

d. **Byte Streams are used for binary data, while Character Streams are used for text data.**

Correct Answer: d

d. **Byte Streams are used for binary data, while Character Streams are used for text data.**

Detailed Solution:

Byte Streams: Handle raw binary data like images or files (InputStream, OutputStream).
Character Streams: Handle text data and support encoding like Unicode (Reader, Writer).
Character Streams are better for text processing, especially when handling international characters.
For a detailed understanding of Java I/O streams, refer to the book *Joy with Java*.

QUESTION 7:

What is the primary purpose of thread synchronization in Java?

a. **To allow multiple threads to execute a method at the same time**

b. **To ensure thread execution follows a specific order**

c. **To prevent rare conditions and ensure data in-consistency**

d. **To prevent race conditions and ensure data consistency**

Correct Answer: d

d. **To prevent race conditions and ensure data consistency**

Detailed Solution:

Thread synchronization is used to control the access of multiple threads to shared resources, ensuring data consistency and preventing race conditions. This is typically done using synchronized methods or blocks.
For more details on thread synchronization, refer to the book *Joy with Java*.

QUESTION 8:

Which stream does Java application uses to only READ data from a source, it may be a file, an array, peripheral device or socket?

a. **InputStream**

b. **OutputStream**

c. **Input/OutputStream**

d. **DataOutputStream**

Correct Answer: a

a. **InputStream**

Detailed Solution:

`InputStream` is used to read data from any source in Java.

QUESTION 9:

Which class in Java is used to create a new directory?

a. **FileReader**

b. **FileWriter**

c. **File**

d. **Directory**

Correct Answer: c

c. **File**

Detailed Solution:

The File class is used to create directories and manipulate files.

QUESTION 10:

What is the primary purpose of input streams in Java?

a. **To write data to a file.**

b. **To read data from a file.**

c. **To delete a file.**

d. **To create directories.**

Correct Answer: b

b. **To read data from a file.**

Detailed Solution:

Input streams are used to read data from a file or another source.

PROGRAMMING IN JAVA

Assignment VIII

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which package must be imported to use AWT components in Java?

a. `java.swing.*`

b. `java.awt.*`

c. `java.gui.*`

d. `java.window.*`

Correct Answer: b

b. java.awt.*

Detailed Solution:

All AWT components such as `Frame`, `Button`, and layout managers are defined in the `java.awt` package.

QUESTION 2:

Which AWT class is used to create a top-level window?

a. `Panel`

b. `Dialog`

c. `Frame`

d. `WindowManager`

Correct Answer: c

c. Frame

Detailed Solution:

The `Frame` class is used to create a top-level window with a title bar, border, and close button.

QUESTION 3:

Which method is used to make an AWT Frame visible?

a. `show()`

b. `display()`

c. `setVisible(true)`

d. `open()`

Correct Answer: c

c. setVisible(true)

Detailed Solution:

The `setVisible(true)` method displays the frame on the screen. Without this call, the window is not shown.

QUESTION 4:

Which layout manager is the default for an AWT `Frame`?

a. `FlowLayout`

b. `GridLayout`

c. `BorderLayout`

d. `CardLayout`

Correct Answer: c

c. `BorderLayout`

Detailed Solution:

An AWT `Frame` uses `BorderLayout` as its default layout manager.

QUESTION 5:

Which method is used to add a component to an AWT container?

a. `insert()`

b. `attach()`

c. `add()`

d. `append()`

Correct Answer: c

c. add()

Detailed Solution:

The `add()` method is used to place components such as buttons or labels into a container.

QUESTION 6:

Which AWT component is used to display non-editable text?

a. `TextField`

b. `TextArea`

c. `Label`

d. `Button`

Correct Answer: c

c. Label

Detailed Solution:

The `Label` component is used to display read-only text in an AWT application.

QUESTION 7:

Which layout manager places components one after another from left to right and wraps to the next line if needed?

a. `GridLayout`

b. `BorderLayout`

c. `FlowLayout`

d. `CardLayout`

Correct Answer: c

c. FlowLayout

Detailed Solution:

`FlowLayout` arranges components from left to right and moves to the next line when space runs out.

QUESTION 8:

Which method is used to enable absolute positioning for components in AWT?

a. `setBounds()`

b. `setLayout(null)`

c. `setPosition()`

d. `setCoordinates()`

Correct Answer: b

b. setLayout(null)

Detailed Solution:

Calling `setLayout(null)` disables layout management and allows manual positioning using `setBounds()`.

QUESTION 9:

Which listener interface is used to handle button click events in AWT?

a. `PrintListener`

b. `KeyListener`

c. `ActionListener`

d. `WindowListener`

Correct Answer: c

c. ActionListener

Detailed Solution:

Button click events in AWT are handled using the `ActionListener` interface.

QUESTION 10:

Which method is used to specify the size of an AWT Frame?

a. `resize()`

b. `setBounds()`

c. `setSize()`

d. `setDimension()`

Correct Answer: c

c. setSize()

Detailed Solution:

The `setSize(width, height)` method defines the size of a frame in pixels.

PROGRAMMING IN JAVA

Assignment 09

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

The container displays a number of components in a row, but adjusts according to the window size.





Which of the following layout manager(s) is most naturally suited for the layout given below?

a. **FlowLayout**

b. **BorderLayout**

c. **GridLayout**

d. **BoxLayout**

Correct Answer: a

a. **FlowLayout**

Detailed Solution:

FlowLayout arranges components left-to-right, wrapping to the next row as needed.

QUESTION 2:

In Java, what is the primary purpose of a layout manager?

a. **To manage memory allocation**

b. **To arrange GUI components within a container**

c. **To handle exception handling**

d. **To control database connections**

Correct Answer: b

b. **To arrange GUI components within a container**

Detailed Solution:

A layout manager in Java is responsible for arranging and positioning GUI components within a container.

QUESTION 3:

Consider the following code.

```
import javax.swing.*;
import java.awt.event.*;
public class NPTEL {
    public static void main(String[] args) {
        JFrame frame = new JFrame("NPTEL Java Course");
        JButton button = new JButton("Click Me");
        button.setBounds(50, 100, 100, 40);
        button.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
            JOptionPane.showMessageDialog(null, "Welcome to the course");
            }
        });
        frame.add(button);
        frame.setSize(300, 200);
        frame.setLayout(null);
        frame.setVisible(true);
    }
}
```

What happens when the button in this Java code snippet is clicked?

a. **The program exits**

b. **A message dialog with the text "Welcome to the course" is displayed**

c. **The button label changes to "Welcome to the course"**

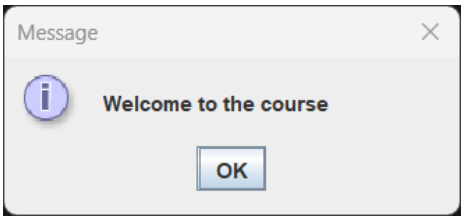
d. **Nothing happens**

Correct Answer: b

b. **A message dialog with the text "Welcome to the course" is displayed**

Detailed Solution:

The code creates a button with label “Click Me” and in the frame titled “NPTEL Java Course”. An action listener is defined that opens a new message dialog with the text “Welcome to the course” when the button is clicked.



QUESTION 4:

Consider the following code.

```
import javax.swing.*;

public class NPTEL extends JFrame {
    JButton button;

    public NPTEL() {
        button = new JButton("Programming in Java");
        add(button);
        setSize(300, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    public static void main(String[] args) {
        new NPTEL();
    }
}
```

What will be the output of the above code?

a. **Creates a JFrame with a JButton labeled "Programming in Java"**

b. **Compiles with errors**

c. **Displays a message dialog**

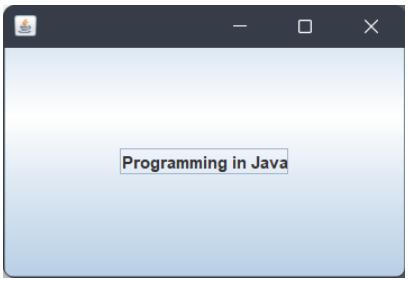
d. **Creates a JPanel with a JButton labeled "Programming in Java"**

Correct Answer: a

a. **Creates a JFrame with a JButton labeled "Programming in Java"**

Detailed Solution:

The code extends `JFrame` and uses the `JButton` class object to create a button with the name of "Programming in Java".



QUESTION 5:

Consider the following code.

```
import javax.swing.*;

public class FrameExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Demo Frame");
        JButton button = new JButton("Click Me");
        // INSERT CODE HERE
        frame.setSize(300, 200);
        frame.setVisible(true);
    }
}
```

What should replace `// INSERT CODE HERE` to create a `JFrame` with a `JButton` labeled `"Click Me"`?

a. **frame.add(button);**

b. **frame.insert(button);**

c. **frame.append(button);**

d. **frame.push(button);**

Correct Answer: a

a. **frame.add(button);**

Detailed Solution:

The `add()` method is used to add components like buttons, text fields, or panels to a `JFrame`.

QUESTION 6:

Identify and correct the error in the following program:

```
import javax.swing.*;
import java.awt.*;

public class PanelExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Panel Example");
        JPanel panel = new JPanel();
        JButton button = new JButton("Submit");
        panel.setFlowLayout(); // ERROR
        panel.add(button);
        frame.add(panel);
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

What should the erroneous line (`panel.setFlowLayout();`) be replaced with?

a. **panel.setLayout(new);**

b. **panel.addFlowLayout();**

c. **panel.appendLayout(new FlowLayout());**

d. **panel.setLayout(new FlowLayout());**

Correct Answer: d

d. **panel.setLayout(new FlowLayout());**

Detailed Solution:

The `setLayout()` method is used to define the layout manager for a panel. `FlowLayout` is the default layout for `JPanel`.

QUESTION 7:

Which method is used to remove a component from a container in AWT?

a. **remove()**

b. **deleteComponent()**

c. **removeComponent()**

d. **destroy()**

Correct Answer: a

a. **remove()**

Detailed Solution:

The remove() method is used to remove a component from a container in AWT. It takes a Component object as an argument and removes it from the container.

QUESTION 8:

Which method is used to handle mouse click events in Java Swing?

a. **mouseClicked()**

b. **keyPressed()**

c. **actionPerformed()**

d. **componentShown()**

Correct Answer: a

a. **mouseClicked()**

Detailed Solution:

The `mouseClicked()` method in the `MouseListener` interface is used to handle mouse click events in Java Swing.

QUESTION 9:

Which Swing component is best suited for displaying a drop-down list of selectable options?

a. **JButton**

b. **JComboBox**

c. **JTextField**

d. **JPanel**

Correct Answer: b

b. **JComboBox**

Detailed Solution:

`JComboBox` is a Swing component that provides a drop-down list from which users can select one option.

QUESTION 10:

Which of the following function is used to specify the layout of a container.

a. **UseLayout()**

b. **setLayout()**

c. **layout()**

d. **DesignLayout()**

Correct Answer: b

b. **setLayout()**

Detailed Solution:

The setLayout() function is used to specify the layout of a container.

PROGRAMMING IN JAVA

Assignment X

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which Java package contains networking-related classes?

a. `java.io`

b. `java.util`

c. `java.net`

d. `java.lang`

Correct Answer: c

c. java.net

Detailed Solution:

All core networking classes in Java, such as `URL`, `Socket`, `InetAddress`, and `ServerSocket`, are part of the `java.net` package.

QUESTION 2:

Which class is used to represent an IP address in Java?

a. `IPAddress`

b. `InetAddress`

c. `NetAddress`

d. `HostAddress`

Correct Answer: b

b. InetAddress

Detailed Solution:

The `InetAddress` class represents an Internet Protocol (IP) address and provides methods to obtain host names and IP addresses.

QUESTION 3:

Which method is used to obtain the IP address from an `InetAddress` object?

a. `getAddress()`

b. `getIP()`

c. `getHostAddress()`

d. `getNetworkAddress()`

Correct Answer: c

c. getHostAddress()

Detailed Solution:

The `getHostAddress()` method returns the IP address in string format.

QUESTION 4:

Which protocol is used by `Socket` in Java for communication?

a. **UDP**

b. **HTTP**

c. **TCP**

d. **FTP**

Correct Answer: c

c. **TCP**

Detailed Solution:

The `Socket` class uses the TCP protocol, which is connection-oriented and reliable.

QUESTION 5:

Which class is used to create a server-side socket in Java?

a. `Socket`

b. `ClientSocket`

c. `ServerSocket`

d. `NetServer`

Correct Answer: c

c. ServerSocket

Detailed Solution:

The `ServerSocket` class waits for client requests and accepts incoming connections.

QUESTION 6:

Which method converts a `URI` object into a `URL` object?

a. `convertURL()`

b. `getURL()`

c. `toURL()`

d. `openURL()`

Correct Answer: c

c. toURL()

Detailed Solution:

The `toURL()` method converts a `URI` into a `URL` obj.

QUESTION 7:

What value does `getPort()` return if no port is specified in the URL?

a. 0

b. 80

c. -1

d. 443

Correct Answer: c

c. -1

Detailed Solution:

If a port number is not explicitly mentioned in the URL, `getPort()` returns `-1`.

QUESTION 8:

Which protocol is connectionless?

a. **TCP**

b. **HTTP**

c. **UDP**

d. **SMTP**

Correct Answer: c

c. **UDP**

Detailed Solution:

UDP is a connectionless protocol that sends data as packets without establishing a connection.

QUESTION 9:

Which method is used by a client to establish a connection with a server?

a. `accept()`

b. `connect()`

c. `listen()`

d. `open()`

Correct Answer: b

b. connect()

Detailed Solution:

A client uses the `connect()` method internally when creating a `Socket` to establish a connection with the server.

QUESTION 10:

Which layer of the OSI model does TCP belong to?

a. **Application layer**

b. **Network layer**

c. **Transport layer**

d. **Data link layer**

Correct Answer: c

c. **Transport layer**

Detailed Solution:

TCP operates at the transport layer and provides reliable, ordered, and error-checked data delivery.

PROGRAMMING IN JAVA

Assignment 11

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

How do you establish a connection to a database using JDBC?

a. **By creating an instance of the Connection interface**

b. **By using the DriverManager.getConnection() method**

c. **By implementing the Connection interface**

d. **By extending the Connection class**

Correct Answer: b

b. **By using the DriverManager.getConnection() method**

Detailed Solution:

To establish a connection to a database using JDBC, you use the DriverManager.getConnection() method. This method takes a JDBC URL, username, and password as parameters and returns a Connection object, which represents a connection to the database. The JDBC URL specifies the database type, location, and other connection details.

QUESTION 2:

Which method executes a simple query and returns a single Result Set object?

a. **executeQuery()**

b. **executeUpdate()**

c. **execute()**

d. **run()**

Correct Answer: a

a. **executeQuery()**

Detailed Solution:

The executeQuery() method is used to execute a simple SQL query that returns a single ResultSet object.

QUESTION 3:

Which of the following ensures that the correct driver is used to access each data source?

a. **java.sql.Manager**

b. **java.sql.DriverManager**

c. **java.sql.Statement**

d. **java.sql.DriverSession**

Correct Answer: b

b. **java.sql.DriverManager**

Detailed Solution:

The DriverManager class ensures that the correct driver is used to access each data source. It manages a list of database drivers and handles establishing connections to the database.

QUESTION 4:

What is the purpose of the ResultSet interface in JDBC?

a. **To store the result of a query**

b. **To execute SQL queries**

c. **To manage database connections**

d. **To update data in the database**

Correct Answer: a

a. **To store the result of a query**

Detailed Solution:

The ResultSet interface in JDBC represents the result set returned by a query. It provides methods to navigate through the result set, retrieve data, and get information about the columns in the result set. The ResultSet interface allows you to process and extract data from a database query.

QUESTION 5:

What does setAutoCommit(false) do?

a. **commits transaction after each query**

b. **explicitly commits transaction**

c. **does not commit transaction automatically after each query**

d. **never commits transaction**

Correct Answer: c

c. **does not commit transaction automatically after each query**

Detailed Solution:

setAutoCommit(false) does not commit transaction automatically after each query. That saves a lot of time of the execution and hence improves performance.

QUESTION 6:

The following is a statement in Java using JDBC.

```
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/nptel","joy","java"
    );
```

Which of the following statement is FALSE?

a. **3306 is the default MySQL port.**

b. **Database name is 'nptel'**

c. **The database server is hosted on IP 127.0.0.1**

d. **Username is 'java' and the password is 'joy'**

Correct Answer: d

d. **Username is 'java' and the password is 'joy'**

Detailed Solution:

In the statement, joy is the username and java is the password. Therefore, statement d, which incorrectly claims that the password for the java user is joy, is false.

QUESTION 7:

What is the correct order to close database resources?

a. **Connection then Statement then ResultSet**

b. **ResultSet then Statement then Connection**

c. **Statement then Connection then ResultSet**

d. **Statement then ResultSet then Connection**

Correct Answer: b

b. **ResultSet then Statement then Connection**

Detailed Solution:

The correct order to close database resources is to first close the ResultSet, then the Statement, and finally the Connection. This ensures that all resources are properly released in the correct sequence.

QUESTION 8:

What is the full form of JDBC?

a. **Java Database Connectivity**

b. **Java Data Code**

c. **Java Data Communication**

d. **Java Development Connectivity**

Correct Answer: a

a. **Java Database Connectivity**

Detailed Solution:

JDBC stands for Java Database Connectivity, a Java API used to connect and interact with relational databases. It provides methods to query and update data in a database.

QUESTION 9:

Fill in the missing code to establish a connection to a MySQL database.

```
import java.sql.*;

public class JDBCExample {
    public static void main(String[] args) {
        try {
            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish Connection
            Connection connection = // INSERT CODE HERE;

            System.out.println("Connected to the database.");
            connection.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

What should replace `// INSERT CODE HERE` ?

a. **DriverManager.connect("mysql:localhost:mydb", "user", "password");**

b. **DriverManager.getConnection("jdbc:mysql://localhost:3306/mydb", "user", "password");**

c. **Connection.get("jdbc:mysql://localhost:3306/mydb", "user", "password");**

d. **Driver.connect("jdbc:mysql://localhost:mydb", "user", "password");**

Correct Answer: b

b. **DriverManager.getConnection("jdbc:mysql://localhost:3306/mydb", "user", "password");**

Detailed Solution:

The `DriverManager.getConnection` method establishes a connection to the specified database URL with the provided username and password.

QUESTION 10:

What is the purpose of the `DriverManager` class in JDBC?

a. **To manage database connections.**

b. **To execute SQL queries.**

c. **To fetch data from a database.**

d. **To represent a database record.**

Correct Answer: a

a. **To manage database connections.**

Detailed Solution:

The `DriverManager` class loads the JDBC drivers and establishes connections to databases.

PROGRAMMING IN JAVA

Assignment XII

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which JDBC object is used to iterate over records returned by a SELECT query?

a. `Statement`

b. `Connection`

c. `ResultSet`

d. `DriverManager`

Correct Answer: c

c. ResultSet

Detailed Solution:

A `ResultSet` object stores and allows traversal of rows returned by a SELECT query.

QUESTION 2:

Which method is used to display a Swing `JFrame` on the screen?

a. `display()`

b. `setVisible(true)`

c. `showFrame()`

d. `open()`

Correct Answer: b

b. setVisible(true)

Detailed Solution:

A Swing frame becomes visible only after calling `setVisible(true)`.

QUESTION 3:

What is the index of the first element in a Java array?

a. 1

b. -1

c. 0

d. Depends on array size

Correct Answer: c

c. 0

Detailed Solution:

Java arrays are zero-indexed.

QUESTION 4:

Consider the following program: (note space appears only one at a time)

```
public class Question {
    public static void main(String[] args) {

        String str = "NPTEL - Programming in JAVA - JAN 2026";

        System.out.println(str.length());
    }
}
```

a. 38

b. 39

c. 40

d. 41

Correct Answer: a

a. 38

Detailed Solution:

The provided Java program calculates and prints the length of the string `str`, which contains the text "NPTEL - Programming in JAVA - JAN 2026".
The output of this program will be: 38

QUESTION 5:

Which keyword is used to handle exceptions in Java?

a. `error`

b. `handle`

c. `catch`

d. `throwable`

Correct Answer: c

c. catch

Detailed Solution:

The `catch` block handles exceptions thrown in the `try` block.

QUESTION 6:

What is the output of the following program?

```
public class Test {
    public static void aMethod() throws Exception {
        try {
            throw new Exception();
        } finally {
            System.out.print("finally ");
        }
    }

    public static void main(String args[]) {
        try {
            aMethod();
        } catch (Exception e) {
            System.out.print("exception ");
        }
        System.out.print("finished ");
    }
}
```

a. **finally**

b. **exception finished**

c. **finally exception finished**

d. **Compilation fails**

Correct Answer: c

c. **finally exception finished**

Detailed Solution:

The program is syntactically correct and here for two try blocks, there is one catch block.
Here's the step-by-step explanation:

I. The `main` method calls `aMethod()`, which throws an `Exception`.
II. Inside `aMethod()`, an `Exception` is thrown in the `try` block.
III. The `finally` block is always executed, regardless of whether an exception is thrown or not. In this case, it prints "finally ".
IV. Since `aMethod()` throws an `Exception`, control moves to the `catch` block in `main`.
V. The `catch` block prints "exception ".
VI. After the `try-catch` block in `main`, "finished " is printed.
VII. Therefore, the complete output is "finally exception finished".

QUESTION 7:

What is the output of the following program?

```
class Program {
    public static void main(String[] args) {
        int counter = 10;
        do {
            System.out.print(2 / counter);
            counter--;
        } while (counter > 2);
    }
}
```

a. 00000012

b. 00000000

c. 10011001

d. 12211221

Correct Answer: b

b. 00000000

Detailed Solution:

The provided program uses a do-while loop to iterate from counter = 10 down to counter = 3. During each iteration, it prints the result of 2 / counter.

1. counter = 10: 2 / 10 results in 0 (integer division).
2. counter = 9: 2 / 9 results in 0.
3. counter = 8: 2 / 8 results in 0.
4. counter = 7: 2 / 7 results in 0.
5. counter = 6: 2 / 6 results in 0.
6. counter = 5: 2 / 5 results in 0.
7. counter = 4: 2 / 4 results in 0.
8. When counter = 3: 2 / 3 results in 0.

After counter becomes 2, the loop condition counter > 2 fails, and the loop terminates.
Therefore, the output of the program is 00000000.

QUESTION 8:

What should be the value of X and Y for the output of the below program to be 36?

```
public class Question {
    public static void main(String[] args) {
        int X = 4;
        int Y = 5;
        int sum = 0;
        for (int i = 0; i < X; i++) {
            for (int j = i; j < Y; j++) {
                sum = sum + j;
            }
        }
        System.out.print(sum);
    }
}
```

a. X = 6 and Y = 5

b. X = 2 and Y = 7

c. X = 1 and Y = 10

d. X = 4 and Y = 5

Correct Answer: d

d. X = 4 and Y = 5

Detailed Solution:

The following is the output of the program for different values of X and Y:

a. *when* X = 6 and Y = 5 *output is* 40

b. *when* X = 2 and Y = 7 *output is* 42

c. *when* X = 1 and Y = 10 *output is* 45

d. *when* X = 4 and Y = 5 *output is* 36

QUESTION 9:

Which of the following options correctly initializes the elements of the numbers array with values 1, 2, 3, 4, and 5?

```
public class NPTEL {
    public static void main(String[] args) {
        int[] numbers = new int[5];
        // #1 : Missing code block
        System.out.println("First element: " + numbers[0]);
    }
}
```

a. **numbers = {1, 2, 3, 4, 5};**

b.
```
for (int i = 1; i < numbers.length; i++) {
    numbers[i] = i;
}
```

c. **numbers[] = {1, 2, 3, 4, 5};**

d. **numbers = new int[]{1, 2, 3, 4, 5};**

Correct Answer: d

d. **numbers = new int[]{1, 2, 3, 4, 5};**

Detailed Solution:

`numbers = new int[]{1, 2, 3, 4, 5};` is the correct answer because it initializes the `numbers` array with values 1, 2, 3, 4, and 5 using array initializer syntax.

QUESTION 10:

Which of the following statements are correct and would NOT cause a compilation error?

```
 I.   float[] = new float(3);
II.   float f1[] = new float[];
III.  float[] f2 = new float[3];
IV.   float f3[] = new float[3];
 V.   float f4[]= { 1.0f, 2.0f, 2.0f };
VI.   float f5[] = new float[] { 1.0f, 2.0f, 3.0f};
```

a. iii, iv, v, vi

b. i, ii, iii, iv

c. ii, iii, v, vi

d. i, ii, iv, vi

Correct Answer: a

a. iii, iv, v, vi

Detailed Solution:

Option iii, iv, v and vi are syntactically correct for declaration of an array.